Estimation of High-Resolution Fractional Tree Cover Using Landsat Time-Series Observations

Jilong Chen, Yang Liu, Ronggao Liu, and Xuexin Wei

***Abstract*— High-resolution fractional tree cover mapping allows for the representation of spatial details of tree distribution and contributes to forest ecosystem monitoring and modeling. However, the estimation of tree cover at high resolution is challenging, especially in sparsely tree-covered and mountainous areas interfered with background signals (e.g., soil, grass, and water) and terrain shadow. This article presents a fractional tree cover estimation algorithm from Landsat time-series observations at 30-m resolution, with the effects of background and terrain shadow reduced. The seasonal profiles of vegetation index and surface reflectance were constructed from multiyear Landsat data. Three phenological metrics were extracted from the seasonal profiles as input features for tree cover estimation. The training data were collected from the European Space Agency (ESA) WorldCover product and used to calibrate a feedforward neural network model to predict cover fractions. This algorithm extracted the tree cover of major forest types, including boreal, temperate, tropical dry, and moist forests. It also captured the sparse tree cover in areas containing mixtures of tree crowns and grass, bare soil, crop, impervious surface, and water. In two dense montane forest areas, the tree cover fractions on shady and sunny slopes were estimated consistently. The estimation results were evaluated through the reference samples generated from Google submeter-resolution image classification. The values of R-squared (R^2), root-mean-square error (RMSE), and mean absolute error (MAE) reached 0.78, 15.71%, and 11.09%, respectively. The proposed algorithm can be applied to monitor tree cover in spatially fragmented forest areas and sparse forests.**

***Index Terms*— Forest, fractional tree cover, high resolution, Landsat, time-series observations.**

I. Introduction

FORESTS provide essential ecosystem services for human well-being [1]. As a vital terrestrial carbon sink and a natural climate regulator, forests store large amounts of carbon from the atmosphere and serve as a nature-based solution to achieve the climate target of preventing global temperatures from increasing more than 1.5 °C [2], [3]. Forests provide wildlife with food and shelter also contribute to the preservation of biodiversity [4]. They also aid in soil and water conservation, wind protection, noise reduction, and air purification [5], [6]. Forest cover maps are fundamental for carbon cycle and climate modeling [7], [8], [9], wildlife habitat conservation [10], and sustainable forest management [11].

Satellite remote sensing provides repeated spectral measurements of the Earth's surface for forest monitoring. Categorical forest/land cover and fractional tree cover are the two main means of forest mapping based on satellite data [12], [13], [14], [15]. In categorical land cover maps, those pixels are classified as forests when the tree cover reaches a certain percentage [16], [17], [18]. This discrete classification method inadequately characterizes within-class heterogeneity of forest density [19]. Fractional tree cover can characterize the variations in forest density with the proportion covered by trees in pixel [20]. This fractional estimation method improves tree cover monitoring in forest-to-nonforest transition zones and fragmented areas and captures forest change at the subpixel level [21], [22].

Many fractional tree cover estimation methods have been proposed, such as empirical regression, spectral unmixing, and machine learning models [23], [24], [25]. Machine learning methods have been widely used to predict tree cover fractions because of their superior performance in fitting multivariate nonlinear relationships [26], [27]. This approach includes the procedures of training sample collection, feature extraction, model calibration, and prediction. Input features are vital to distinguish land cover types and improve the model performance [28]. Spectral reflectance, vegetation indices, and their descriptive statistics as well as harmonic regression coefficients derived from time-series data are usually extracted as input features for tree cover estimation [28], [29], [30]. Regression methods, such as random forest [31], regression tree [32], gradient boosting regression [33], convolutional neural network (CNN) [34], and empirical regression [35], were applied to establish relationships between extracted features and tree cover fractions. Several global fractional tree cover maps have been produced from satellite data at a spatial resolution ranging from tens to thousands of meters. These include the Advanced Very High Resolution Radiometer (AVHRR) and Moderate Resolution Imaging Spectroradiometer (MODIS) vegetation continuous field (VCF) datasets [36], [37], the Copernicus Global Land Service Land Cover (CGLS-LC100) dataset [38], the 30-m Global Forest Cover Change Tree Cover

Manuscript received 7 May 2023; revised 23 August 2023; accepted 26 September 2023. Date of publication 17 October 2023; date of current version 23 October 2023. This work was supported in part by the National Key Research and Development Program of China under Grant 2019YFA0606600, in part by the National Natural Science Foundation of China under Grant 42161144001, in part by the Youth Innovation Promotion Association of Chinese Academy of Sciences under Grant 2019056, and in part by the Youth Project of Innovation LREIS under Grant YPI008. *(Corresponding author: Yang Liu.)*

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Digital Object Identifier 10.1109/TGRS.2023.3323641

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(GFCC30TC) dataset [39], and the Global Forest Change (GFC) dataset [40]. These products generally perform well in dense forest areas, while notable differences are presented in areas with low crown cover, such as sparse woody cover in the Sahel, and fragmented forests in urban and agricultural areas [41], [42], [43].

High-resolution fractional tree cover estimation can improve the description of tree distribution in low crown cover areas and tree cover variability [44], [45]. However, high-resolution tree cover mapping from remote sensing data shows relatively high uncertainties. As the spatial resolution increases from hundreds or thousands of meters to tens of meters, some factors that can be ignored at low resolutions would become the main signals, such as backgrounds (e.g., soil, grass, and water) and topographic effects. These factors may interfere with the extraction of tree cover from high-resolution satellite observations, especially in sparse tree-covered areas and mountainous areas. In sparsely tree-covered areas, the contribution of the background to the spectral reflectance of satellite observations cannot be ignored and may even exceed that of tree crowns, making it difficult to extract fractional tree cover in areas such as savanna, urban, and cropland [25], [46]. In mountainous areas, terrain shadow, which is rarely considered in low-resolution satellite images, will significantly change the spectral reflectance of high-resolution images and interfere with the crown signal [47], [48]. It was reported that the two existing ten-meter-resolution global tree cover products, GFC and GFCC30TC, showed relatively high uncertainties in dryland biomes and mountain areas [49], [50], [51], [52], [53], [54]. This is probably related to the use of spectral reflectance as input features for tree cover extraction, which is sensitive to background mixture and terrain shadow.

Vegetation indices can strengthen vegetation information through band combination and reduce the impacts of background and topographic effects [55], whereas it may be difficult to distinguish between trees and dense herbaceous vegetation only based on vegetation indices. The shapes of greenness seasonal curves have significant differences between the two vegetation types [56]. Trees show a greenness plateau during the vegetation maturity period, while herbaceous vegetation shows a parabolic shape with a downward opening. The use of time-series vegetation indices can improve the separability between the two types [28] and weaken the influences of background signal and terrain shadow.

This article aims to develop a high-resolution fractional tree cover estimation algorithm based on satellite time-series data. Here, tree cover refers to the maximum woody vegetation cover when leaves are at full growth during a year. We try to reduce the interference of background signal and terrain shadow on tree cover extraction. Seasonal profiles of vegetation index and surface reflectance were constructed from multiyear Landsat observations and used to extract several phenological metrics with high separability of trees. Fractional tree cover was estimated using a calibrated neural network at 30-m resolution. The proposed algorithm was tested in regions covered with main forest types, sparsely tree-covered areas with different backgrounds and mountain areas. The accuracy was evaluated with the reference samples generated from Google image classification.

TABLE I
LANDSAT DATA USED TO ESTIMATE FRACTIONAL TREE COVER

Sensor	Landsat 8		Landsat 5	
Bands	Band 4 (Red)	Band 5 (Near infrared)	Band 3 (Red)	Band 4 (Near infrared)
Spatial resolution (m)	30			
Path/Row	044/019, 044/020, 045/019, 045/020, 046/019; 116/029, 117/029; 127/048, 127/049, 128/048; 231/069, 232/068, 232/069		114/026, 115/026; 166/068, 167/068, 167/069; 230/078; 019/031, 020/030, 020/031; 035/041; 231/076, 231/077; 148/037, 149/037	
Acquisition date	2017-2021		1998-2002	

II. DATA AND METHOD

A. Data and Preprocessing

1) Landsat Surface Reflectance: The surface reflectance in red and near-infrared (NIR) bands of Landsat 8 images from 2017 to 2021 was used to develop the fractional tree cover estimation algorithm (Table I). To compare with two global 30-m tree cover products that are both available in the 2000 epoch, the surface reflectance in red and NIR bands of Landsat 5 images from 1998 to 2002 was also utilized to retrieve tree cover fractions. These data collected in Landsat Collection 2 Level-2 Tier 1 datasets were obtained from the Google Earth Engine (GEE) platform [57]. In these datasets, the Landsat 5 and 8 surface reflectance data were created with the Landsat Ecosystem Disturbance Adaptive Processing System (LEDAPS) algorithm and the Land Surface Reflectance Code (LaSRC) algorithm, respectively [58]. All land surface reflectance for visible and NIR bands were atmospherically corrected. The quality layer was also produced based on the C Function of Mask (CFMask) algorithm [59]. In this study, the quality layer was utilized to mask cloud shadow, water, and snow pixels, and the surface reflectance in red and NIR bands was then used to calculate Normalized Difference Vegetation Index (NDVI) and generate phenological metrics.

2) Tree Cover Products for Comparison: The estimated fractional tree cover results were compared with the GFC and GFCC30TC products, which provide global percent tree cover maps at 30-m resolution. The GFC tree cover map was generated from Landsat data based on a bagged decision tree method [40]. A series of per band metrics was employed, including selected percentile values of spectral reflectance, mean reflectance values for selected percentiles, and slope of linear regression of reflectance versus image date. The overall accuracy of the GFC tree cover map was reported to be above 90% at the national scale [60]. GFCC30TC includes global tree cover maps for the years circa-2000, 2005, 2010, and 2015 by rescaling the tree cover layer of the MODIS VCF dataset [39]. A piecewise linear function model was applied

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TABLE II
GOOGLE IMAGES USED FOR REFERENCE SAMPLE GENERATION

Dominant land cover	Longitude	Latitude	Area (km^2)	Acquisition date	Reference sample number
Boreal forests	115.69°W-115.66°W	58.39°N-58.41°N	8.38	2019-05-26	2300
Temperate forests	127.86°E-127.88°E	44.56°N-44.58°N	5.33	2018-05-19	1443
Tropical dry forests	104.40°E-104.42°E	16.78°N-16.80°N	3.70	2020-12-27	1023
Tropical moist forests	63.50°W-63.48°W	12.65°S-12.63°S	5.48	2021-07-09	1482
Urban	83.25°W-83.19°W	42.40°N-42.44°N	45.73	2019-06-08	12636
Cropland	103.98°E-104.00°E	16.74°N-16.72°N	4.19	2018-12-05	1155

to the Landsat data to predict tree cover fractions at 30-m resolution. The overall accuracy was reported to be 91% [49].

3) Reference Sample for Validation: Field measurements can provide "ground truth" for validation of estimates, but it is hard to collect adequate ground measurements due to high costs in terms of time and labor [61]. Moreover, there is generally a mismatch between field data collection area and remote sensing pixel size [27]. Submeter satellite imagery can capture the fine distribution of tree crown. The fractional tree cover can be estimated by calculating the percentage of tree cover pixels in target grids. This method has been widely used for reference sample generation [53]. In this study, we selected six submeter Google images, including four major forest areas, an urban area, and a cropland area (Table II). These areas were mostly selected from the demonstration areas, while the urban area was adjacent to the demonstration urban site. The urban was selected with a larger area to generate more reference samples of fractional tree cover with intermediate values for model validation. To reduce the possible errors caused by the measurement date and the climate, we used the images acquired closest to the vigorous growing season from 2017 to 2021. These images were classified into two classes ("Tree = 1" and "Nontree = 0") using CNN [62] (Fig. 1). The 927 polygons were manually labeled as samples totally, including 462 tree polygons and 465 nontree polygons. These samples were randomly split into two sets, 70% as the training set and the remaining 30% as the test set. A basic CNN model including three convolution layers and four dropout layers was built using Tensorflow 2.12.0. The rectified linear unit (ReLU) algorithm was applied as the activation function [63]. The digital number (DN) values of the three bands of Google images were normalized and used for model inputs. The classification accuracy of all images was above 0.95. In view of the positional misalignment between Google images and Landsat images, Google image classification results were aggregated into 60-m grids to generate 60-m resolution fractional tree cover maps as the reference samples. The 20 039 samples were totally collected for validation. The estimated 30-m fractional tree cover results were also resampled into 60-m grids to compare with the reference samples.

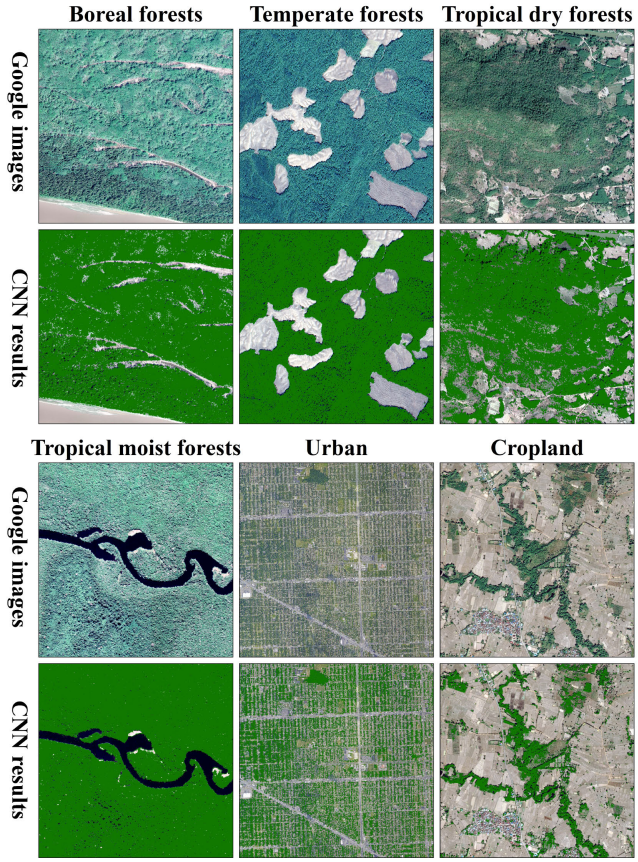


Fig. 1. Classification results of six submeter Google images used for reference sample generation based on CNN method. The extracted trees are shown in green in the second and fourth rows.

B. Method

Fractional tree cover was estimated based on a trained feed-forward neural network (FNN) and three phenological metrics extracted from Landsat time-series data (Fig. 2). Seasonal profiles of NDVI and surface reflectance were constructed from five-year Landsat images. Three phenological metrics were then extracted from the constructed seasonal data. The training samples were created from the European Space Agency (ESA) WorldCover product. An FNN model was calibrated with the training samples and the phenological metrics and applied to estimate tree cover fractions from Landsat data.

1) Seasonal Profile Construction: The seasonal profiles were constructed from five-year Landsat NDVI and surface reflectance in the red band. Cloud, cloud shadow, water, and snow were masked based on the quality layer contained in the Landsat surface reflectance dataset. NDVI was calculated with the surface reflectance of the red and NIR bands. The maximum value compositing (MVC) method was used to construct seasonal profiles in view of that this method can reduce directional reflectance and off-nadir viewing effects and minimize cloud contamination, shadow effects, and aerosol and water vapor effects [64], [65]. The largest NDVI value every eight days was chosen to represent the growth curves of vegetation, which helped to exclude possible cloud and shadow residues. When the absolute differences between an NDVI value and two adjacent NDVI values were both greater

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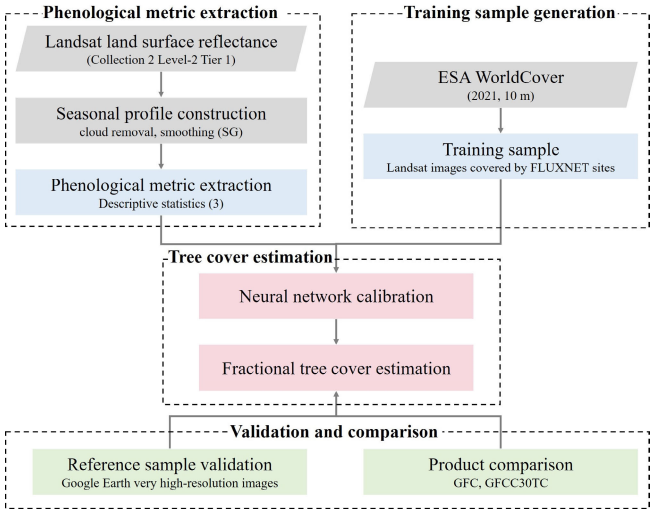


Fig. 2. Workflow for fractional tree cover estimates and quality assessment.

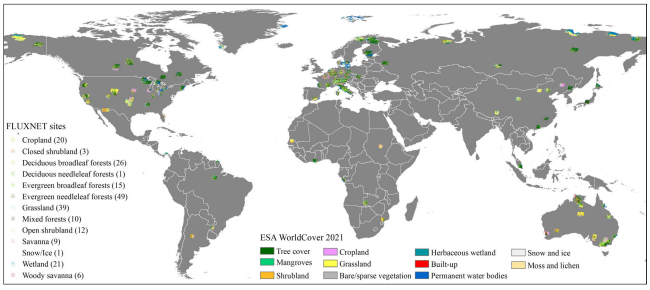


Fig. 3. Distribution of ESA WorldCover for generating training samples around 212 FLUXNET sites.

than 0.3, the NDVI value was removed to reduce the impact of outliers. Gaps in the constructed NDVI curves were filled and smoothed by the Savitzky Golay filter [66]. Furthermore, the red band reflectance corresponding to the selected NDVI was preserved to construct the seasonal profiles of vegetation. It is a sensitive measure of both spatial and temporal variations in vegetation photosynthetic activity. Reflectance in the red band generally shows a negative correlation with tree cover [67], [68]. And its time series of forests also exists a low-value plateau that does not find for herbaceous vegetation [69], [70]. The features derived from the red band reflectance values during the whole growing season may enhance tree cover information.

2) Phenological Metric Extraction: Three phenological metrics were calculated from the constructed seasonal data as input features. We constructed the eight-day composite NDVI series to represent the seasonal profiles of vegetations. In ideal conditions, an NDVI sequence of a year should contain 46 valid NDVI values. Considering the difference of NDVI plateau length of trees in various climate zones, we selected consecutive 11, 13, 16, and 36 largest NDVI values to calculate the phenological metrics. These numbers of NDVI were chosen to represent the variability of the length of the NDVI plateau for boreal forest (88 days), temperate forest (104 days), tropical dry forest (128 days), and tropical moist forest (288 days). Due to different shapes of the NDVI curve for trees and herbaceous vegetation, two phenological metrics named the mean ($NDVI_{Mean}$) and the bias ($NDVI_{Bias}$) of NDVI of the growing season were extracted to discriminate these two vegetation types [56]. For each pixel, $NDVI_{Mean}$ was the mean of the selected NDVI values, and $NDVI_{Bias}$ was calculated by averaging the difference between the maximum NDVI value and each selected NDVI value. Additionally, the surface reflectance of the red band corresponding to the chosen NDVI values was calculated similar to that of $NDVI_{Bias}$ and referred to as another phenological metric named Red_{Bias}.

3) Training Sample Generation: The training samples were generated from the ESA WorldCover 2021 product around 212 FLUXNET sites (Fig. 3). This product provides a global 10-m resolution land cover map based on Sentinel-2 multispectral image data and Sentinel-1 C-band synthetic aperture radar (SAR) data [71]. The producer's accuracy and user's accuracy of the tree cover class reached 91.9% and 80.0%. These sites represent diverse ecosystems classes, including evergreen broadleaf forest (15), deciduous broadleaf forest (26), evergreen needleleaf forest (49), deciduous needleleaf forest (1), mixed forest (10), woody savanna (6), savanna (9), closed shrubland (3), open shrubland (12), grassland (39), cropland (20), wetland (21), and snow/ice (1) [72]. For each FLUXNET site, a Landsat image-sized polygon was chosen to extract the ESA WorldCover 2021 land cover map. The values of tree cover pixels in the WorldCover image were set to 100%, and that of other land cover classes were set to 0%. Then, all the 10-m pixels were aggregated to 300-m grids to generate the fractional tree cover map. The 72 605 029 grids were obtained initially. In order to ensure the quality of training samples, those grids with less than 23 valid NDVI values were excluded. Finally, a total of 35 275 595 grids were collected as the training samples. To exclude possible errors, the tree cover fractions for grids less than 5% were set to 0%, and the tree cover fractions for grids greater than 95% with the annual maximum aggregated Landsat NDVI above 0.85 were set to 100%.

4) Neural Network Calibration and Fractional Tree Cover Estimation: An FNN was applied to estimate fractional tree cover. FNN is an artificial neural network where information moves in only one direction, from the input nodes through the hidden layer to the output nodes [73]. Three phenological metrics ($NDVI_{Mean}$, $NDVI_{Bias}$, and Red_{Bias}) derived from five-year Landsat surface reflectance data were aggregated to 300-m grids and set as three neurons of the input layer. The hidden layer contained five layers containing 400, 200, 100, 50, and 25 neurons. The ReLU algorithm was applied as the activation function [63]. The tree cover fraction was the only neuron in the output layer. The neural network model was calibrated with the training samples. Since the continuous variable is robust to the change in spatial resolution of the imagery [39], [74], [75], [76], the fit neural network model was applied to estimate fractional tree cover from the 30-m Landsat phenological metrics. Tree cover fractions of pixels with the maximum NDVI values in five composition years less than 0.35 was set to 0.

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5) Validation and Comparison: The fractional tree cover estimation results were assessed with the reference samples derived from the classified submeter-resolution imageries of Google Earth at 60-m resolution in view of the positional misalignment between Google images and Landsat images. The R-square (R^2) and slope were regressed from a fit ordinary least-squares regression line to evaluate the accuracy of the estimated results. The root-mean-square error (RMSE) and the mean absolute error (MAE) were also calculated between the estimated tree cover fractions and the reference samples. In addition, the results were compared with two existing 30-m resolution tree cover products in areas of grassland, miombo woodland, cropland, urban, mangroves, and montane forests to qualitatively illustrate the applicability of the proposed algorithm.

III. Demonstration Areas

Four example areas of main forest ecoregions were chosen to illustrate the performance of the fractional tree cover estimation algorithm in different forest types (Table III). These areas lie in boreal forests (Area I, 116.17°W–115.15°W, 58.06°N–58.59°N), temperate forests (Area II, 127.32°E–128.31°E, 44.32°N–45.02°N), tropical dry forests (Area III, 103.72°E–104.61°E, 16.33°N–17.19°N), and tropical moist forests (Area IV, 64.29°W–63.32°W, 13.32°S–12.41°S). The map of Earth's terrestrial ecoregions was used as a reference for forest types [77]. Area I is located in Boreal Plains Forests ecoregion in central Canada (3473.03 km^2). The mixed coniferous and deciduous forests are the dominant vegetation in this area. The main species are quaking aspen and balsam poplar mixed with black and white spruce. Area II is located in Manchurian Mixed Forests ecoregion in northeastern China (6060.93 km^2). Mongolian oak, birch, maple, and poplar are the main deciduous broadleaf tree species in this area. Area III is located in Central Indochina Dry Forests ecoregion in eastern Thailand (8997.64 km^2). The area is covered with an open woodland dominated by Dipterocarpus species. Area IV lies in northeastern Bolivia (10 579.10 km^2) covered with evergreen broadleaf forests.

Five sparsely tree-covered areas containing mixtures of grass, soil, crop, impervious surface, water, and a low proportion of tree crowns, and two montane forest areas affected by terrain shadow were selected as examples to demonstrate the performance of the proposed algorithm (Table III). Site a (337.34 km^2) is a grassland area characterized by mixed grass and tree cover in southeastern Russia (131.97°E–132.23°E, 48.50°–48.66°N). Site b (15.37 km^2) is an area of dry miombo woodland in northern Mozambique (37.02°–37.06°E, 12.41°–12.38°S) influenced by the soil and dry herbaceous vegetation backgrounds. Sites c (24.67 km^2) and d (2.51 km^2) are cropland in northern Argentina (64.11°–64.06°W, 25.97°–25.92°S) and urban in the eastern United States (83.26°–83.24°W, 42.38°–42.39°N) areas, which contain a low proportion of tree cover. Site e (18.65 km^2) is located in the Northern Mesoamerican Pacific Mangroves ecoregion in western Mexico (110.56°–110.52°W, 27.38°–27.42°N). Remote sensing imagery over this area contains mixtures of mangroves and water. Sites f (10.30 km^2) and g (21.24 km^2) are two mountain areas affected by terrain shadow. The two sites lie in northwestern Argentina (65.17°–65.13°W, 23.87°–23.83°S) and northern India (75.37°E–75.41°E, 33.55°N–33.58°N), respectively.

TABLE III
Location of Demonstration Areas. Four Example Areas Are Covered With Different Forest Types, and Seven Typical Sites Are Influenced by Background Mixture and Terrain Shadow

Name	Dominant land cover	Longitude	Latitude	Area (km^2)	Ecoregion
Area I	Boreal forests	116.17°W-115.15°W	58.06°N-58.59°N	3473.03	Mid-Canada Boreal Plains Forests
Area II	Temperate forests	127.32°E-128.31°E	44.32°N-45.02°N	6060.93	Manchurian Mixed Forests
Area III	Tropical dry forests	103.72°E-104.61°E	16.33°N-17.19°N	8997.64	Central Indochina Dry Forests
Area IV	Tropical moist forests	64.29°W-63.32°W	13.32°S-12.41°S	10579.10	Madeira-Tapajós Moist Forests
Site a	Grassland	131.97°E-132.23°E	48.50°N-48.66°N	337.34	Ussuri Broadleaf and Mixed Forests
Site b	Miombo woodland	37.02°E-37.06°E	12.41°S-12.38°S	15.37	Dry Miombo Woodlands
Site c	Cropland	64.11°W-64.06°W	25.97°S-25.92°S	24.67	Dry Chaco
Site d	Urban	83.26°W-83.24°W	42.38°N-42.39°N	2.51	Southern Great Lakes Forests
Site e	Mangroves	110.56°W-110.52°W	27.38°N-27.42°N	18.65	Northern Mesoamerican Pacific Mangroves
Site f	Mountain forests	65.17°W-65.13°W	23.87°S-23.83°S	10.30	Southern Andean Yungas
Site g	Mountain forests	75.37°E-75.41°E	33.55°N-33.58°N	21.24	Western Himalayan Broadleaf Forests

IV. Results

A. NDVI Seasonal Profiles of Vegetation

Time-series NDVI for the forests and adjacent grassland sites was used to illustrate the seasonal curve of greenness for forests and herbaceous vegetation. The NDVI seasonal profiles were constructed from Landsat images from 2017 to 2021 for six major forest types, including deciduous broadleaf forests in China (108.53°E, 35.76°N), evergreen broadleaf forests in Argentina (56.46°W, −28.49°S), deciduous coniferous forests in Mongolia (100.99°E, 50.48°N), evergreen coniferous forests in the United States (107.54°W, 37.34°N), monsoon forests in Myanmar (94.85°E, 19.30°N), and tropical savanna in Mozambique (34.70°E, 17.69°S). According to very high-resolution imageries from Google Earth, a nearby grassland site was chosen for each forest site. As shown in Fig. 4, the constructed Landsat NDVI curves can capture different shapes for the two vegetation types. During the vegetation maturity period, forests have a stable greenness plateau with high NDVI values around 0.8 or above. Grassland shows a parabolic shape with an opening downward and the NDVI values vary notably. Although the NDVI values of the grassland nearby the evergreen broadleaf forests are mostly

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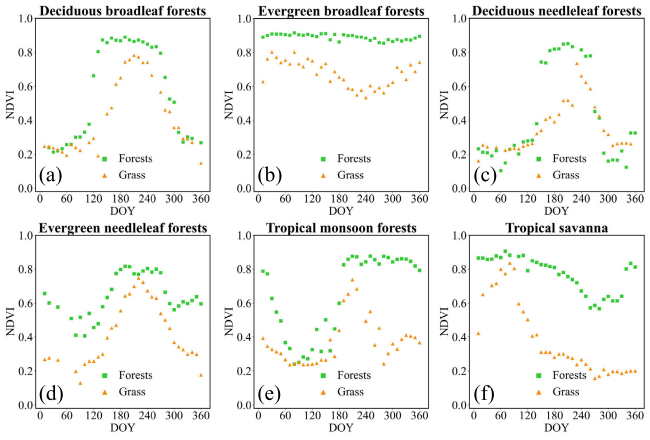


Fig. 4. Constructed NDVI seasonal profiles for various forest types and adjacent grass sites. These include sites located in (a) deciduous broadleaf forests in China (108.53°E, 35.76°N), (b) evergreen broadleaf forests in Argentina (56.46°W, −28.49°S), (c) deciduous coniferous forests in Mongolia (100.99°E, 50.48°N), (d) evergreen coniferous forests in the United States (107.54°W, 37.34°N), (e) tropical monsoon forests in Myanmar (94.85°E, 19.30°N) and (f) tropical savanna in Mozambique (34.70°E, 17.69°S). The NDVI time-series data were generated from 2017 to 2021 Landsat images.

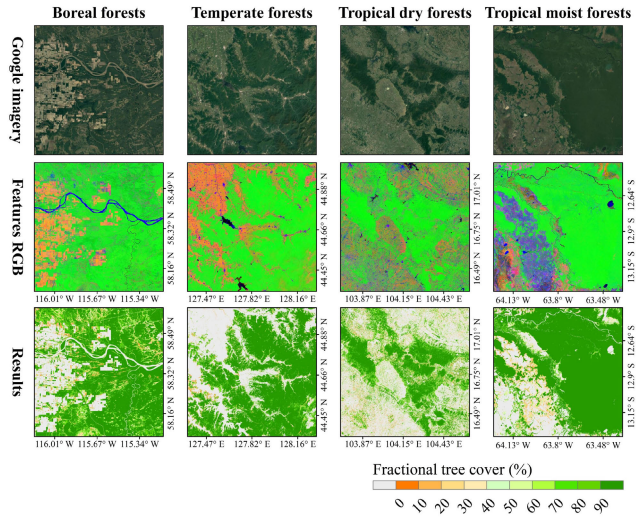


Fig. 5. Distribution of phenological metrics and estimated fractional tree cover for the main forest types. The first and third rows show Google Earth's very high-resolution imageries and the estimated fractional tree cover results in the 2020 epoch. The second row is the false color composite images of three phenological metrics. The red, green, and blue use $NDVI_{Bias}$, $NDVI_{Mean}$, and Red_{Bias} for the second row, respectively.

above 0.5, notable differences were observed between them [Fig. 4(b)]. Due to different NDVI shapes, the $NDVI_{Mean}$ of forests should be larger than that of herbaceous vegetation, and the $NDVI_{Bias}$ should be smaller than the latter.

B. Distribution of Phenological Metrics and Estimated Fractional Tree Cover

Fig. 5 presents the distribution of phenological metrics and estimated fractional tree cover derived from Landsat images from 2017 to 2021 for four areas covered with different forest types. The very high-resolution images of Google Earth were also shown as a reference. $NDVI_{Bias}$, $NDVI_{Mean}$, and Red_{Bias} were used to composite the false color images (the second row). These composited images show notable contrast of color and brightness. Forests generally present green. Pixels with high tree cover proportions have a bright green color, and the color of sparse trees is light. Cropland and grassland are orange-red, yellow, or purple, and construction land and bare land appear blue and red. The significant color contrasts suggest that the phenological metrics have great separability for trees and other land cover components. The proposed algorithm extracted the distribution of forests with reasonable tree cover values referencing the Google true-color imageries. The estimated cover fraction reaches 80% or even more than 90% for dense forests, while it ranges from 10% to 70% in the forest edges and patches.

C. Comparison With Existing Tree Cover Product

The estimated fractional tree cover results from Landsat images from 1998 to 2002 were compared with two existing tree cover products, GFC and GFCC30TC in circa 2000. The results were compared in five selected areas containing mixtures of tree crowns and different backgrounds, including grass, soil, crop, impervious surface, and water body, and two montane forest areas that were affected by terrain shadow. The very high-resolution imageries of Google Earth from around 2000 were displayed as a reference.

The fractional tree cover results for five areas containing mixtures of tree crowns and different backgrounds are shown in Fig. 6 (the second row). In the grassland, the tree cover of a large number of tree patches with high-density crown cover is estimated above 70%. For the scattered trees mixed with bare soil and dry grass in the miombo woodland, less than 30% of tree cover is retrieved. The trees linearly distributed around the cropland are successfully captured, with the estimates ranging from 10% to 30%. In the urban area, the algorithm extracts the trees that form closed crowns and scattered trees with about 20% cover between buildings. Mangrove forests mixed with water background are also detected, with the tree cover below 60%. For the other two fractional tree cover products, GFC and GFCC30TC misclassify grass into trees in the grassland area, and the estimates are even more than 60%. These two products also underestimate high crown cover and overestimate low crown cover in soil background areas. GFC has a similar performance in urban areas with our results, while it cannot capture the tree crowns of the cropland and mangrove areas. GFCC30TC allows for the representation of tree cover of the urban and mangrove areas, but the values are underestimated compared to Google imageries.

Fig. 7 presents the performance of our algorithm and two existing 30-m resolution tree cover products in two mountain areas. The results of our algorithm show no obvious difference on the shady and sunny slopes, and the tree cover fractions of most pixels are above 90%. GFC and GFCC30TC are notably influenced by the terrain. These two products both underestimate the tree crowns on the sunny slopes. GFCC30TC also underestimates the results on the shady slopes. To further demonstrate the performance of the proposed algorithm, the histograms of tree cover percentage for the shady and sunny slopes (65.17°–65.13°W, 23.87°–23.83°S) are presented in

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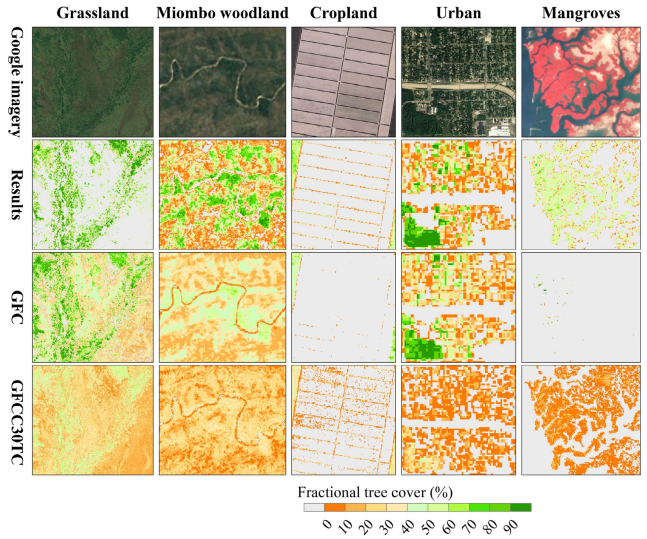


Fig. 6. Comparison of fractional tree cover in sparsely tree-covered areas. The first row exhibits the Google imageries and the others display the fractional tree cover of the estimated results (the second row), GFC (the third row), and GFCC30TC (the fourth row) in the 2000 epoch. For the mangrove forest area, a false color image composited from the NIR, red, and green bands of Landsat in the growing season was presented, in which the mangroves appeared red.

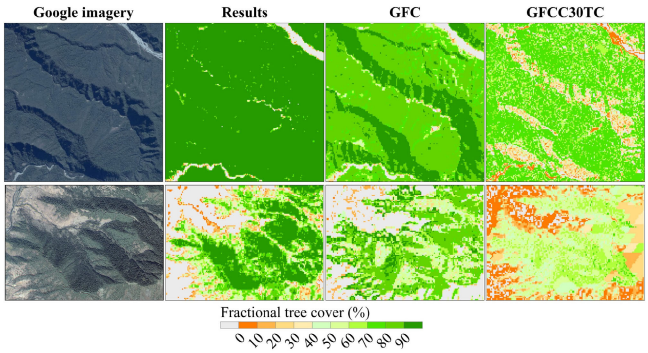


Fig. 7. Comparison of fractional tree cover in montane forest areas. The first column exhibits Google's imageries and the others display the fractional tree cover of the estimated results (second column), GFC (third column), and GFCC30TC (fourth column) in the circa 2000.

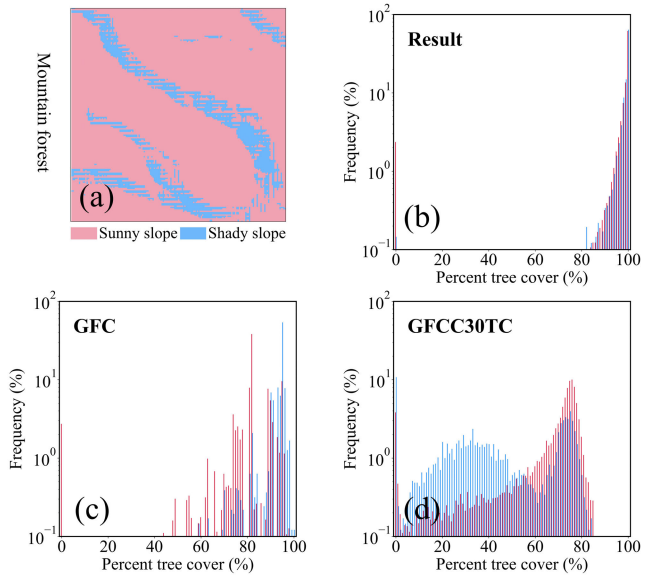


Fig. 8. Histograms of the tree cover fractions on shady and sunny slopes: (a) is the distribution of shady and sunny slopes and (b)–(d) are the statistics of the fractional tree cover for the estimated result, GFC, and GFCC30TC.

Fig. 8. The Shuttle Radar Topography Mission (SRTM) digital elevation data [78] and the Sun azimuth and Sun elevation of Landsat 5 images were used to calculate the area of the shady and sunny slopes. The comparison results show that the consistency of the estimated tree cover on shady and sunny slopes is improved using the developed algorithm, and the cover fractions of most pixels range from 80% to 100%, which indicates that the influence of terrain shadow is reduced. GFC performs well on shady slopes with most estimated values reaching 95%, but the tree cover fractions on the sunny slope are between 70% and 90%, with a more than 20% underestimation. For the GFCC30TC product, 72% of the pixel values in the shaded area are below 70%, while 65% of the pixel values in the shined area are above 70%.

A deciduous broadleaf forest site (123.89°E, 41.10°N) in the mountain of Northeast China was selected to demonstrate the feasibility of the proposed algorithm for extracting montane forests. Two pixels of shady and sunny slopes completely covered by dense forests were chosen according to very high-resolution imageries of Google Earth. The constructed seasonal profiles of NDVI and corresponding surface reflectance in the red and NIR bands were extracted from Landsat images from 2017 to 2021 for the two pixels (Fig. 9). The results show that the reflectance of the NIR and red bands on the sunny slope is much higher than that of the shady slope, and the NIR band is more sensitive to the topographic effects than the red band [Fig. 9(c) and (d)]. In contrast, the NDVI values of shady and sunny slopes in the growing season are close [Fig. 9(b)]. Thus, NDVI and other features derived from time-series NDVI data will be less affected by terrain shadow than spectral reflectance, which helps the proposed algorithm to reduce the influence of mountain shadow in forest cover extraction in mountain areas.

D. Accuracy Assessment

The accuracy of the estimation results was evaluated with 20 039 reference samples derived from Google submeter-resolution image classification across four areas covering main forest types, an urban area, and a cropland area. The scatterplot is shown in Fig. 10. The points are mainly spread close to the 1:1 line. The R^2 between the reference samples and estimated fractions is 0.78, and the slope value is 0.98. The RMSE and the MAE are 15.71% and 11.09%, respectively. The validation results are influenced by the following factors. The fractional tree cover was derived from five-year Landsat observations, and the validation samples were obtained from Google images on a single day during this five-year period. Seasonal and interannual changes in tree cover may introduce errors into the validation results. In addition, the proposed method seems to underestimate the sparse forested with tree cover of less than 40%, which may be explained by the fact that the submeter-resolution imagery

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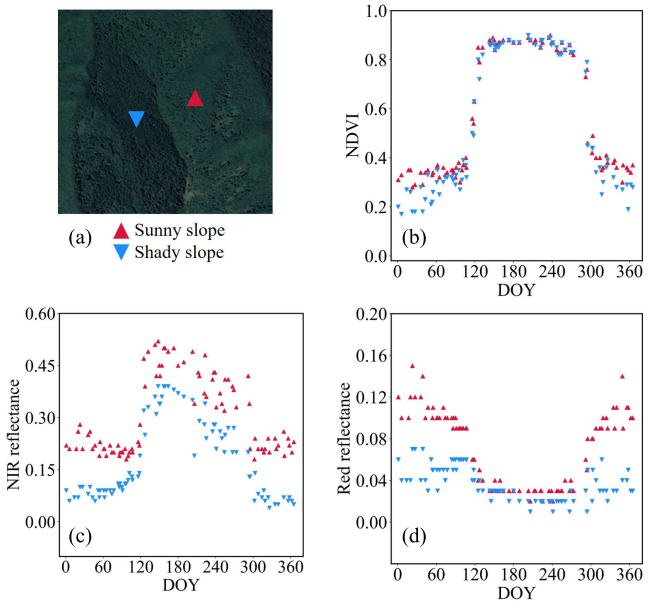


Fig. 9. (a) Seasonal profiles of NDVI and surface reflectance in the NIR and red bands of forests on shady and sunny slopes. The two sites with a close crown cover lie in the mountain area of Northeast China (123.89°E, 41.10°N). (b)–(d) Time-series data for NDVI and surface reflectance in the NIR and red bands, respectively.

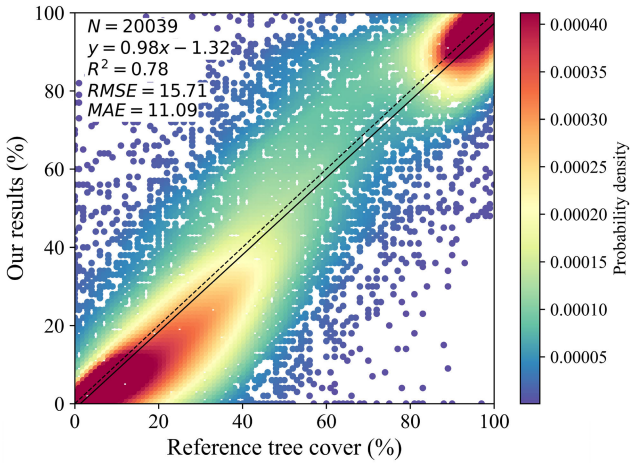


Fig. 10. Scatterplot of estimated fractional tree cover versus reference samples derived from Google image classification.

can capture more small woody plants that is omitted in tens of meters resolution satellite imagery.

V. Discussion

This article presents a high-resolution fractional tree cover estimation algorithm based on Landsat time-series observations. The results demonstrate that the proposed method can extract the fractional tree cover for the main forest types, and it can capture the tree cover of sparsely tree-covered and mountainous areas, with the reduction of the interference of background and terrain shadow. The retrieval of biophysical variables from high-resolution satellite data is always affected by background signals, such as soil, water, and grass [46], [79], [80]. Here, we consider and exploit time-series differences across land cover types, which helps to differentiate tree cover from background and capture the distribution of tree crowns in sparsely tree-covered areas. Additionally, this algorithm is insensitive to topographic effects. Vegetation indices based on reflectance ratios such as NDVI can reduce the impact of terrain shadow [55]. In view of different imaging times of satellite images and the sun–target–sensor geometry, the shaded area in one image may be the shined area in another image. The multiyear maximum NDVI compositing method may extract the NDVI values that are less affected by the terrain shadow. The phenological metrics that condense time-series NDVI data will further weaken this impact. Thus, the proposed algorithm can improve the tree cover estimation on shady and sunny slopes. In mountainous areas that are severely affected by terrain shadow, such as the northwest margin of the Himalayas, the effects of huge terrain shadow cannot be eliminated by the phenological metrics, which may introduce notable uncertainties in tree cover estimation. The performance of our algorithm depends on the number of valid satellite observations. It is difficult to obtain dense time-series data from Landsat satellites due to a 16-day revisiting time and cloud cover, especially for the period before Landsat 8 launched [58]. In this study, seasonal curves of NDVI were constructed from five-year Landsat observations. This method might result in a few potential issues. The maximum NDVI per eight days selected from five-year Landsat data may distort the original vegetation growth curves and extend the vegetation greenness period, due to the advanced start of the growing season (SOS) or delayed end of the growing season (EOS). These issues possibly result in the misclassification of woody and herbaceous vegetation. And interannual land cover change and forest disturbance influenced by human activities or climate change will be ignored [81]. The high-frequency and high-resolution satellite observations will help to acquire the seasonal data for tree cover estimation. For example, the Copernicus Sentinel-2 mission provides satellite observations at high spatial resolution (10–60 m) and short revisit time (about five days). Combining the Landsat and Sentinel-2 observations will contribute to improving tree cover estimation.

The proposed algorithm may be influenced by the quality of the training data source. In this article, the ESA WorldCover product was used to generate training samples. This product shows relatively good quality for tree cover identification with 91.9% producer's accuracy and 80.0% user's accuracy of the tree cover class. To reduce the impact of possible classification errors, we generated more than 35 million training samples around 212 FLUXNET sites worldwide. Furthermore, to evaluate the influence of training sample data source and resolution, we generated the 30-m fractional tree cover samples utilizing the submeter-resolution image classification to create tree cover maps and compare with the original estimation results. For the demonstration area of tropical dry forests, four Google images were chosen and classified into two classes ("Tree = 1" and "Nontree = 0") using a basic CNN model (Fig. 11) [62]. The classification results were then aggregated into 30-m grids as the training samples for 30-m fractional tree cover estimation. Fig. 12 presents the estimated tree cover with the training samples derived from WorldCover and Google images. It shows that these two

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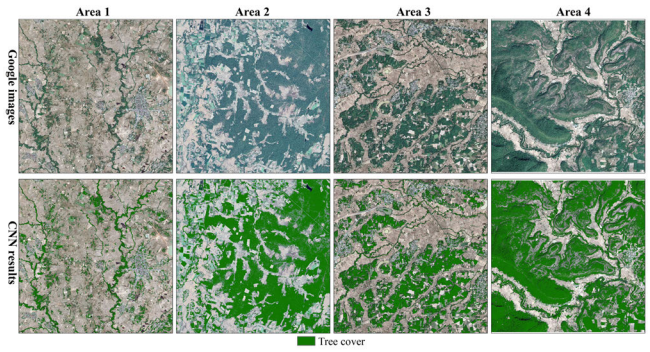


Fig. 11. Classification results of four submeter Google images used for training sample generation based on CNN method. The extracted trees are shown in green in the second row.

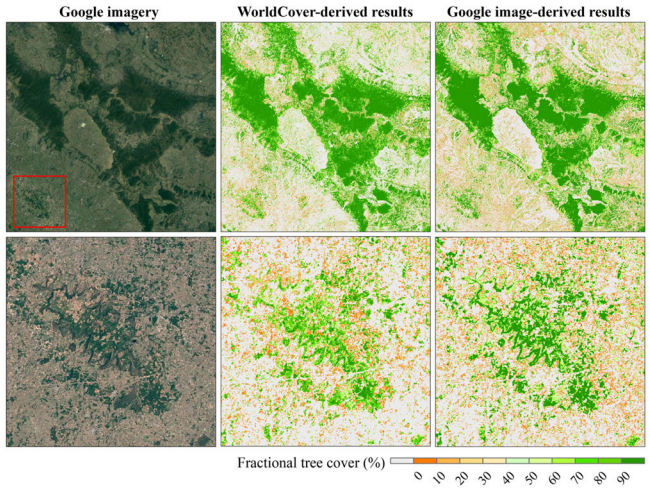


Fig. 12. Fractional tree cover maps extracted using the reference samples derived from the WorldCover land cover product and Google image classification. The red box is the area of the zoomed results in the second row.

methods extracted similar spatial distribution patterns of tree cover. The Google image-derived results are about 5%–10% higher than the WorldCover-derived results in sparsely tree-covered areas. This may be explained by the fact that the submeter-resolution imagery can capture more small trees that are omitted in the ESA WorldCover product. It is possible to obtain higher quality training data based on submeter image classification. However, it should be noted that the spatial extent of submeter Google imagery is limited, and the processing of these images is time-consuming and labor-intensive. The algorithm proposed in this article makes full use of the existing high-resolution data products and can perform 30-m fractional tree cover estimation in large areas in the absence of Google Images data.

Our algorithm may be influenced by the saturate issue of NDVI. NDVI will be saturated when the amount of vegetation leaf pigments reaches a critical point [82], [83], [84]. NDVI saturation occurs mostly in dense forests, such as lush tropical rainforests and temperate forests. The NDVI saturation phenomenon may make the algorithm unable to capture the subtle cover change of dense forests and introduce some uncertainty to the high values of tree cover estimates. But in this case, tree cover is usually close to 100%, and these uncertainties should be limited. For nonforests, NDVI saturation may occur in some areas of dense herbaceous vegetation during the most vigorous growing season, which is mainly the densely planted cropland. In these areas, the NDVI seasonal curve will show a plateau in the middle of the growing season. However, the length of this NDVI plateau of herbaceous vegetation should be much shorter than that of forests. Thus, the extracted input features based on the NDVI values within the whole growing season are still capable of differentiating forests from herbaceous vegetation.

VI. Conclusion

In this article, a high-resolution fractional tree cover estimation algorithm based on Landsat time-series observations was developed. The seasonal curves of NDVI and surface reflectance in the red band were constructed from multiyear Landsat observations and used to extract three phenological metrics that show high separability of trees. Training data were created by aggregating the ESA WorldCover land cover product. An FNN model was calibrated and used for fractional tree cover fraction estimation at 30-m resolution. This algorithm was tested in several forest ecoregions, sparsely tree-covered areas and mountain forest areas. The estimated results showed that the proposed algorithm could extract the tree cover for major forest types and reduce the interferences of background effect and terrain shadow. When comparing the estimated fractional tree cover with 20 039 reference samples generated from classified submeter-resolution imageries of Google Earth, the R^2, MAE, and RMSE values reached 0.78, 11.09%, and 15.71%, respectively. The proposed algorithm can be applied to estimate tree cover fraction for complex land surfaces and montane forests from high-resolution satellite data, contributing to the monitoring of the distribution and change of forest cover.

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